Swarna Venkata Sai Rohit

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Education

Indian Institute of Technology Bhilai (IIT Bhilai) — **July 2023 – May 2027**
B.Tech. in Electrical Engineering — **CGPA: 8.86 | Best SGPA: 9.47**

Experience

Superfone | *Software Engineering Intern — Backend, APIs & Integrations [Certificate]* — **May 2026 – Jul 2026**

- Built and published the Superfone iPaaS app on Pabbly Connect and Zapier end-to-end (Node.js / TypeScript / Express) — WABA v1 template messaging, v2 campaign broadcast API, and v3 deferred-enrichment media webhooks, with dynamic dropdowns, x-api-key auth, and header-based idempotency; passed both partner app-directory reviews.
- Built server-side "Invoke Superfone Variables" auto-filling templates from CRM/org data at send time and double-debit-safe wallet handling (MongoDB / PostgreSQL / Redis); audited the superfone-api codebase and scoped a Zoho + HubSpot integration, flagging blockers pre-build.

IIT Bombay — TIH-IoT | *Software Engineering Intern — Real-Time Systems [Certificate]* — **Dec 2024 – Oct 2025**
AR-Based Tool Condition Monitoring — real-time ML fault detection with AR visualization and haptic alerts.

- Built an async, event-driven edge pipeline on Raspberry Pi ingesting 1,000+ IoT sensor readings/sec at sub-100ms latency via non-blocking WebSocket I/O (timeout-wrapped `recv()`, retry-with-backoff reconnection); deployed a VAE-based 1D-CNN classifier for real-time tool-wear detection (fresh / worn / critical).
- Designed a hybrid REST (control plane — auth, channel config, endpoint discovery) + WebSocket/MQTT Pub/Sub (data plane) architecture streaming to CNC machines, HoloLens AR, and ESP32 haptic devices; ran fault-tolerant Linux systemd services at 99.5% uptime with PyTest coverage.

IIT Bhilai — IBITF | *Software Engineering Intern — Platform Development [Certificate]* — **Jun 2025 – Sep 2025**
Unified Role-Based Digital Platform for Campus Operations.

- Built and shipped a full-stack platform (React + TypeScript, FastAPI, PostgreSQL) for 1,000+ students and 70+ faculty across academic, mess, and healthcare workflows; REST APIs with async/await, Pydantic validation, and pooling.
- Implemented server-side RBAC across 10+ roles with authorization middleware; built modular React + TypeScript UI components with configurable theming, cutting development time by 40%.

DRDO, RCI Hyderabad | *Research Intern — Embedded Systems [Certificate]* — **May 2025 – Jun 2025**

- Developed Python parsers that decode raw hexadecimal OBC telemetry into structured JSON, optimized with bitwise operations and NumPy; validated correctness under a senior scientist for mission-critical data ingestion.

Projects

QuantyX — Real-Time Trading Platform — Live | GitHub 2025

- Architected a 3-service event-driven backend (API Gateway, Execution Service, Event Service) in an NPM-workspaces monorepo — Redis Pub/Sub as command bus decouples intake from execution, returning API responses in <10ms while a worker HMAC-SHA256-signs orders on Binance Testnet.
- Built a WebSocket fan-out layer (`Map<userId, Set<WebSocket>>`) pushing order-fill updates with zero polling; Next.js 14 + TypeScript frontend renders flicker-free TradingView charts via ref-based updates, backed by PostgreSQL + Prisma, Zod, JWT, Docker.

Quantum — LLM-Powered AI Workspace & Conversational Agent — Live Demo | GitHub 2025

- Built a full-stack AI workspace (MERN) orchestrating Gemini 2.5 Flash across 8 modules — multi-turn chat with PDF/DOCX/TXT ingestion, an NLP Insight Engine rendering entities/relationships/sentiment as interactive 3D graphs (Plotly.js), an AI Image Generation module (from NeuraPix), and an Executive Assistant generating strategy reports.
- Engineered retry-with-exponential-backoff (3 attempts, auto Gemini 2.5→1.5-Flash fallback) for AI-call resilience; secured the platform with Passport.js OAuth 2.0 (Google/GitHub), dual-token JWT (15-min access + 15-day HttpOnly refresh), and Helmet/CORS hardening.

SentinelAI — Real-Time CCTV Anomaly & Crime Detection — GitHub 2025

- Engineered a real-time CCTV pipeline in Python — sampled live video frames, extracted 512-D ViT-based CLIP embeddings over a rolling 32-frame buffer for temporal context, and ran DSANet for behavioral anomaly scoring in parallel with YOLOv8 for weapon/firearm detection, triggering a danger event on either signal.
- On a danger event, classified the flagged frame into specific crime categories via a pretrained Hugging Face CCTV-crime classifier, logged event metadata & annotated frames to SQLite, and surfaced real-time alerts on a FastAPI + React dashboard.

Additional: CraterX — lunar crater detection (computer vision, 1.3M+ craters) GitHub | Traffic Demand Prediction — ML on tabular data (LightGBM / XGBoost, 5-fold CV, $R^2 \approx 0.95$) GitHub | NeuraPix — AI image-generation platformGitHub.

Skills

Languages: Python, C++, C, TypeScript / JavaScript, SQL
Frontend: React + TypeScript, Next.js, Tailwind CSS
Backend, APIs & Databases: Node.js, Express.js, FastAPI, RESTful APIs, WebSockets, Microservices, event-driven / async architectures, OAuth 2.0 / JWT (Passport.js), Pydantic; PostgreSQL, MongoDB, MySQL, Redis, Prisma
CS Fundamentals: DSA (400+ — LeetCode / GeeksforGeeks / CodeChef, Best Rank 714), OOP, Operating Systems, Computer Networks (TCP/IP), DBMS, Probability & Statistics
DevOps & Tools: Git / GitHub, CI/CD (GitHub Actions), Docker, Linux (systemd), Unit & Integration Testing (PyTest), Railway, iPaaS (Pabbly Connect, Zapier), WhatsApp Business API (WABA)
AI / ML: LLMs, LLM APIs, Machine Learning, Deep Learning, Transformers, Vision Transformers (ViT), OpenCV, YOLOv8 (applied in projects)

Awards & Honors

- **Flipkart GRiD 8.0** — Qualified to Round 3 (Pan-India) | **Amazon ML Summer School** — Selected, 2026
- **Conference Publication & Presentation (Springer Nature)** — Intl. VETOMAC Conference, IIT Guwahati | Publication
- **Young Researcher Award** — IIT Bhilai Alumni Association | Certificate
- **TIH-IoT Chanakya Fellowship** — IIT Bombay | **IBITF Technology Innovation Fellowship** — IIT Bhilai
- **ATF Stage-I Qualified** (Competitive Programming) — Top 4,000 / 20,000+ candidates | Certificate
- **Website Development Lead** — Student Mentorship Programme, IIT Bhilai | Website